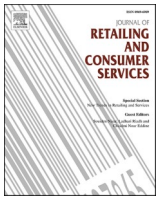
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The diffusion of mobile payments: Profiling the adopters and non-adopters, Roger's way

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ABSTRACT

This study aims to establish various market segments based on consumers' attitudinal elements influencing the adoption of mobile payments (m-payments). It also examines the best discriminators between identified clusters of m-payment users. Three hundred and sixty m-payment users participated in the study. Cluster and discriminant analyses were employed for the analysis. The findings suggest that two segments are established, and strategies can be developed accordingly. The study provides marketers with numerous touchpoints to propagate the adoption of m-payment apps. It also comprehensively explains m-payment users in developing economies by superimposing the cohorts to Roger's diffusion of innovation model.

1. Introduction

Today, the global economy has become interconnected, and so has the demand for an effective and efficient global payment solutions (Bezovski, 2016). With the rise of new payment technologies, the point-of-sale and retail industries are witnessing an upsurge in product demand. Also, the demand for global payment solutions has compelled technology firms, mobile manufacturers, and bankers alike to scramble to provide effective and efficient solutions to customers. Mobile payment solutions (MPS) can address this issue. Mobile payment can be defined as the payment made for a product or service through a portable electronic device such as a tablet or cell phone (Wu et al., 2021; Sankaran and Chakraborty, 2021). Thus, we have mobile wallets and on-demand apps coming to market almost every other day. On their end, consumers are becoming more demanding in terms of personalized and interactive shopping experiences. To keep pace with the rapid changes in payment technology, merchants must first understand the driving factors. For instance, in the financial year (FY) 2022, India's m-payments through BHIM (Bharat Interface for Money) reported approximately INR 85 billion in monthly (January 2022) transaction value. More than 71 billion digital payments were recorded in the financial year (Statista, 2022). Given these stupendous numbers, there is no doubt that m-payments have boomed in India, wherein consumers are using apps for all purposes, from mobile phone account recharges to utility payments to retail transactions (both in-store and online). Moreover, owing to the ongoing Covid-19 pandemic, every form of cashless payment has witnessed a phenomenal rise in usage and demand.

The younger generation is possibly the largest cohort in adopting online and cashless payments (Zhang et al., 2017). However, a better understanding of both motivators and deterrents to m-payment adoption among this cohort could assist service practitioners, and researchers develop appropriate promotional strategies to make m-payments broadly acceptable to a larger group of potential adopters (Wei et al., 2021). Nevertheless, these 'potential adopters' must be 'technology--ready.' According to Wiese and Humbani (2020), technology readiness is critical in understanding users' attitudes towards m-payment apps. Earlier, Guhr et al. (2013) opined that a standard approach might not help examine how users perceive and adopt a new technology per se. This observation is significant as we must remember that m-payments are beyond borders; thus, marketers need to deal with consumers who naturally differ from one another (Bhatnagar and Ghose, 2004).

Quality of life has significantly improved with the use of mobile payments, and continuance intention is impacted by the performance of the service and the effort required to use the service with ease (Purohit et al., 2022). Sankaran and Chakraborty (2021) opined that factors like handling money, ease of use, and convenience motivate consumers to use mobile payments. Consumers also tended to oppose, reject or postpone the use of m-payments during the pandemic owing to the risk and

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traditional barriers (Talwar et al., 2021a,b). Kar (2021) examined the usage satisfaction model for digital services and established that factors like usefulness, trust, credibility, and information privacy lead to first-time usage and increased m-payments. According to Wu et al. (2021), trust is critical in facilitating usage intention toward m-payments. Yong et al. (2021) emphasized the impact of mobility, risk, and cost on user intentions to adopt mobile payments. Mindfulness, perceived ease of use, perceived usefulness, subjective norms, and attitude play a critical role in m-payment usage intention (Flavián et al., 2020). Subjective norm is the degree of individual attention influenced by other societal members' opinions while taking a particular decision (Ajzen and Fishbein, 1980). Attitude can be defined as the beliefs that a person has related to the results that will be offered by the application of certain behaviors and his evaluation of possible outcomes (Ajzen and Fishbein, 1980). Technology readiness index (TRI) constructs aided the perceived usefulness and ease of use in influencing attitudes towards m-payments during the COVID-19 pandemic (Rafdinal and Senalasari, 2021). Market segmentation may be the appropriate answer to identify these varying needs of consumers, as it would help focus on the utmost fruitful segments (Asmi et al., 2016). Furthermore, Wiese et al. (2017) opined that clustering-derived segments might be pronounced intelligent behavior, serving as a foundation for developing and effectively implementing target marketing. Clustering-derived segments are used to identify viable market segments. Because clustered segments can be described in terms of measured behavior, marketers can develop targeting strategies that are specifically tailored to a specific segment or segment (Wiese et al., 2017).

Therefore, the first and foremost step in market segmentation is identifying variables based on which a homogenous cohort can be characterized. Herein, segmentation can be based on readily identifiable variables like demographics, geographies, and social-economic backgrounds. However, a critical challenge in identifying these homogenous segments is that the segments in themselves may not necessarily be demarcated, owing to ever-changing and complex consumer behavior. Thus, the segmentation criteria should include psychographic and behavioral variables that help identify homogenous cohorts, depicting essential differences in their attitudes towards m-payment apps.

Studies in the past have attempted to 'segment' markets to understand mobile phone users. (Kımıloğlu et al., 2010) segmented mobile phone users based on their behavioral attributes. Zhu et al. (2009) segmented users based on lifestyle factors that, in turn, influence user preferences for product attributes. Goneos-Malka et al. (2014) segmented mobile phone users grounded on the aspects of mobile phones, while Chawla and Joshi (2017) focused on mobile banking (mBanking) adoption for segmenting mobile phone users. From these studies, it is evident that none focused on segmenting m-payment users based on their readiness and acceptance (Humbani and Weise, 2019). The current study attempts to bridge the research gap using Roger's Diffusion of Innovation (DoI) model and classify the users based on their attitudinal differences by effectively using hierarchical and k-means clustering classification methods. The current study aims to identify the factors influencing m-payment adoption alongside profiling the users based on extracted factors.

Further, it categorizes factors that discriminate between the user categories, if any. Finally, because mobile wallets are a technology-driven innovation platform, this research looks to examine if the recognized aspects and linkages from the earlier works stand ground in the case of m-payments—the current study pioneers in applying Roger's diffusion of innovation model to identify and segment m-payment users. In a nutshell, this study aims to address the following research questions:

RQ 1: What are the factors influencing m-payment adoption?
RQ 2: Can m-payment users be profiled based on factors influencing m-payment adoption, as identified in RQ1?
RQ 3: Which factors discriminate between the clusters identified in RQ2?

The following section provides a literature review of the variables influencing m-payment users' attitudes and the models. Next, the methodology section describes our study's design, including sampling, measures, and data analysis techniques. The next section deals with the results, followed by a discussion. Finally, it outlines the implications and possible future research directions.

2. Theoretical background and literature review

2.1. The TAM and UTAUT models

The technology acceptance model (TAM) is primarily acknowledged as the utmost accurate, cost-effective, and robust framework for determining how people accept and use technology (Pavlou, 2003; Davis, 1989). The framework is founded on the theory of reasoned action (TRA), which holds that a person's behavior intention leads to actual behavior, which is influenced by subjective norms and attitude, both of which influence individual beliefs (Ajzen and Fishbein, 1980). Online banking acceptance is based on six aspects: perceived usefulness (PU), perceived ease of use (PEOU), perceived enjoyment, knowledge about online banking, security and privacy, and internet connection quality (Pikkarainen et al., 2004). Subjective norms and enabling situations were considered external factors, as suggested by Teo et al. (2008). Singh and Sinha (2020) examined the positive effect of trust in explaining the impact of perceived usefulness on a merchant's intentions to use mobile payment technology. Peer-to-peer mobile payment adoption is significantly impacted by perceived usefulness (Kalinic et al., 2019). Lee et al. (2019) studied the reciprocal relationship between retailers and user perception of platform-based mobile payment. Researchers are increasingly extending the TAM with more variables, as it is the most effective instrument for investigating usage-associated issues (Shin, 2009). Munoz-Leiva et al. (2017) established a TAM incorporating the innovation diffusion theory, perceived risk, and trust in the typical TAM model.

Venkatesh et al. (2003) projected the unified theory of acceptance and use of technology (UTAUT) model to elucidate user intentions towards using an information system and their ensuing behavior. According to the model, four key components (performance expectancy, effort expectancy, social influence, and facilitating conditions) are critical determinants of behavioral intention and, eventually, user behavior. Shin (2009) added security, trust, social influence, and self-efficacy to the UTAUT paradigm. UTAUT has been proposed as an extended TAM and effectively explains technological acceptance (Shin, 2009). TAM and UTAUT have been used to elucidate consumers' intents and acceptance of several technologies, including e-commerce (Zhang et al., 2006), bank technology (Wan et al., 2005), internet banking (Akinci et al., 2004), mobile banking (Lee et al., 2010; Shin, 2009), online learning (Cheung and Lee, 2011) and mobile coupons (Liu et al., 2015; Revels et al., 2010).

2.2. Rogers' diffusion of innovation model

Diffusion of Innovation (DOI) has been considered a landmark theory because it was designed to understand how a product or service originates, gains momentum, and diffuses across society (Johnson et al., 2018; Rogers, 2003). In addition, researchers have used DOI to examine consumers' adoption behavior in different contexts, such as online shopping (Bigne-Alcaniz et al., 2008) and mobile commerce and mobile banking (Van der Boor et al., 2014). Diffusion is a process by which an innovation is spread among members of a social system over time and through specific channels (Rogers, 2003). Diffusion happens due to a mix of: (a) individuals' motivation to lessen personal uncertainty when confronted with new information; b) Individuals want to react to their views of what others are thinking and doing; and (c) broad social pressure to follow in the footsteps of others. Significantly, depending on one's level of awareness and appreciation of innovation, one's

requirements or motives will vary (timeliness in adoption). The first to adopt (innovators) do so because of novelty, whereas the second (early adopters, including opinion leaders) do so based on an assessment of the innovation's merits. The great majority of those who adopt are doing so because of others. This idea may aid in categorizing consumers depending on their amount of innovation and attitudes toward a particular phenomenon. For example, social media engagement is critical to enhancing user enhancement for mobile payment services (Grover and Kar, 2020).

2.3. Mobile payment and its adoption

Mobile payment refers to using mobile devices to pay for goods and services, such as wireless handsets, personal digital assistants, radio-frequency devices (RFIDs), and near-field communication-based devices (Moghavvemi et al., 2021; Chen and Nath, 2008; Dahlberg et al., 2008). A qualitative text mining using reviews from various mobile payment apps explored themes like convenience, security, reliability, transaction speed, and trust that impact the adoption of mobile payments (Verkijika and Neneh, 2021). Banks list m-payment devices, allowing for high-security financial transfers from a payer's bank (or credit card account) to the payee's bank account (Liu and Dewitte, 2021; Ng and Yip, 2010). A cross-country comparison (India and the USA) confirmed that convenience and consumers' willingness to pay were critical drivers for adopting mobile payments (Boden et al., 2020). Mobile payment solutions, despite gaining popularity, face challenges related to their adoption and usage (Kaur et al., 2020a,b). First, numerous MPSs have emerged during digital transformation. As a result, it is critical that users find value in their chosen MPS. If the chosen MPS fails to provide users with the desired value, users may discontinue or switch. Secondly, the use of MPSs is associated with various uncertainties or risks concerning privacy, security, data, and money theft, including risks associated with sending money to the incorrect vendor and paying more money than is required for a transaction (Kaur et al., 2020a,b).

Liao and Yang (2020) opined that the retail payment mechanism helps promote a better consumer experience, therefore promoting the use of mobile payment solutions. They suggested that the role of mobile payment is determined in terms of new retail payment mechanisms that promote a better consumer purchase experience in an online to the offline business environment. Modern consumers do not see a brand in silos; they are likely to have multiple touch-points with a retailer, and the purchase journey between each touch-point and channel must be seamless. According to Park et al. (2019a,b), convenience, economic benefits, and enjoyment may impact the adoption attitudes toward mobile payments. Greenacre and Akbar (2019) opined that payment methods impact how low-income consumers shop and react to market situations. Yan et al. (2021) while studying adoption of Quick response (QR) codes have found convenience and ease as the antecedents. The results showed that the more benefits offered to the users, the more favorable it is for them to adopt QR code m-payment since it can reduce the queue time when consumers want to make payments easily, which allows them to save time and perform other activities.

Mombeuil (2020) has established that each independent variable (relative convenience, relative advantage, perceived privacy, perceived security) positively influences adoption intentions, wherein relative advantage and perceived security are the best predictors of adoption intention based on mobile wallets (Chen et al., 2022).

Since it was first proposed, the TAM has been extensively tested and modified to predict technology adoption behavior, while UTAUT is an extension of TAM (Venkatesh et al., 2003). The UTAUT framework suggests that these four contextual variables: performance expectancy, effort expectancy, social influence and facilitating conditions predict intentions to use (IU), and IU further predicts the actual behavior. The technology adoption model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Oh et al., 2019; Soh et al., 2020) have been studied widely to study the intention to use m-payment. Slade et al. (2015) concluded that PU and PEOU were the most determinant and significant factors influencing the intention to use (IU) of m-payment. Additionally, Shaikh and Karjaluoto (2015) arrived at similar conclusions on mobile banking adoption after examining 55 relevant journal articles.

Recent studies have also applied DOI to m-wallets (Kaur et al., 2020a,b; Johnson et al., 2018; Di Pietro et al., 2015). Consequently, DOI was utilized since the objective was to segment the users adopting mobile payments into adopters and non–adopters. TAM and IDT are both considered to be suitable models for predicting users' intention to adopt new technology. However, TAM is frequently used in studies of mobile services adoption because of its ease of applicability and simplicity (Zhou, 2013; Chong et al., 2012; Zarmpou et al., 2012; Venkatesh and Davis, 2000).

Extant literature leads us to examine the importance of consumer attitudinal factors that motivate the adoption of m-payments and help identify microsegments, if any, based on the factors identified. The authors also endeavor to examine the most critical factors that differentiate the microsegments best to tailor-make strategies to address the specific needs of the identified segments. Table 1 provides an overview of prior research in the area of m-payments.

3. Methods

The target population for the study was Indian millennials (Gen Y) who use mobile payments for purchases. Notably, this Gen Y represents the profile of active m-payment shoppers due to their unique characteristics (Dash et al., 2021). They seek quick and convenient solutions with their product usage and adopt innovations faster. Data were collected online through self-administered questionnaires with the help of Google forms. The questionnaires were sent to 430 targeted participants, out of which 393 responded, indicating a 90% response rate. The area of data collection was a major Indian city and its suburbs, which has a huge Gen Y population. All participants were generation Y m-payment users. Of these responses, only 360 surveys were suitable for post-data screening and cleaning. Only 14 users were between 40 and 45 years of age. Judgmental-cum-purposive sampling method was used for data collection. Therefore, all our participants were those who extensively used m-payments for their purchases.

A pilot pool of questionnaire items was formed after a thorough assessment of the literature on the constructs. Items were adapted from previous studies and modified to fit the m payment context (see Appendix 1). A five-point Likert scale was used to rate these closed-ended items extending from strongly disagree (1) to strongly agree (5). Multiple iterations were done on the pre-existing scale to remove similar meanings, ambiguous statements, and negatively phrased statements to modify the items to the study context. We also added some items for a few constructs to ensure robustness in measuring the construct. Notably, we took precautions by removing responses with missing values.

Five academics and four industry professionals validated the content validity of the survey. Next, we conducted a pretest through focus group discussions (FGDs) with 12 respondents who use m-payments frequently. The FGDs helped us identify five items that were added to the initial pool of thirty-five items taken from pre-established scales. We designed the online survey to capture the demographics of the responders and their attitudes towards m-payments. The respondents reported their age, gender, annual household income (INR lakhs') (1 lakh = 100,000), occupation, and usage category in the survey forms. Following the data collection process, we used the statistical software Statistical Package for Social Sciences (SPSS) 24.0 to analyze and interpret the data. In that sequence, the data was also subjected to a multivariate statistical analysis, including factor analysis, cluster analysis, and discriminant analysis. The details of the respondents are provided in Table 2.

Table 1
Extant research on m-payments and related constructs
TRA - Theory of Reasoned Action, TAM – Technology Acceptance Model, UTAUT – Unified Theory of Acceptance and use of Technology, DOI – Diffusion of Innovation, PU- Perceived Usefulness, PEOU- Perceived Ease of Use, AR - Adoption Readiness, PR - Perceived Risk, PrR – Privacy Risk
* MPS – MPS is an acronym for mobile payment services and is defined as "A mobile payment solution (MPS) is defined as the process of making use of Internet connectivity and mobile devices for processing payments when purchasing goods or services (Di Pietro et al., 2015).

Author (s)	Theoretical Lens	Input variable	Output variable	Methodology	Location	Key findings
Kaur et al. (2020)	DOI	Relative advantage, compatibility, complexity, observability, trialability	intention to use mobile payment, intention to recommend	CB-SEM	Finland	The findings suggest that usage, risk, and value barriers are negatively associated with intentions to use MPSs. In contrast, usage and value barriers negatively affect users' intention to recommend MPSs.
Shankar and Rishi (2020)	Extended TAM, DOI	Self-efficacy, personal innovativeness, subjective norm, trust	PU, PEOU, m-payment adoption	CB-SEM	India	Results indicate that the intention to adopt m-banking leads to m-banking adoption and usage.
Talwar et al. (2020)	Information systems success model, transaction cost economics theory	perceived information quality, perceived success quality, perceived uncertainty, perceived asset specificity	initial trust, PU, PEOU dissatisfaction, continuation intention	CB-SEM	India	Information and service quality positively influence initial trust, which positively correlates with confirmation and PU. Furthermore, a positive relationship between PU and continuation intention was also established.
Liu et al. (2019)	Extended TAM	Perceived mobility, PR, perceived cost	PEOU, PU, and intention to use mobile payment	PLS- SEM	China	Perceived mobility positively impacts PEOU, and PU indirectly impacts adoption intention; PrR and perceived cost negatively affect a user's intention to use mobile payments.
Johnson et al. (2018)	DOI	PrR, Ubiquity, trialability	PEOU, Relative advantage, visibility, perceived security, usage intention	PLS- SEM	USA	PEOU, relative advantage, visibility, and perceived security impact usage, ubiquity, and trialability positively, while PrR negatively impacts perceived security.
Jun et.al (2018)	DOI	Compatibility, simplicity, economic value, switching cost, perceived PrR, and personal innovativeness.	perceived value and intention of continued use	CB- SEM	Korea	All benefit-related variables positively impacted perceived value, wherein compatibility had a significant relationship with perceived value.
Park et al. (2019b)	None	Perceived benefit, PR	Trust on m-payment, PR, Perceived Benefit, and Intention to use	CB-SEM	South Korea	Results supported the negative relationship between PrR, trust, and consumer intention to use m-payment. However, a positive relationship between perceived benefits and trust was found. Trust mediated consumer intention to use m-payment.
Munoz-Leiva et al. (2017)	Extended TAM	Social image	PU, PEOU, attitude, perceived trust, PR, intention to use	CB- SEM	Spain	Attitude determines the intended use of mobile apps, discarding usefulness and risk as factors that directly improve their use
Wu et al. (2017)	Consumer response system and affect heuristic	Positive emotion	PR, PU, acceptance intention	CB-SEM	China	Users' acceptance intention to be related to PR, PU, and positive emotion has a strong negative impact on PR and a positive impact on PU, whereas PU decreases users' perception of risk.
Upadhyay and Jahanyan (2016)	TRA, TAM, DOI	System and task fit, technical perspective, User characteristics, Externalities	PU, PEOU, Intention to use	CB-SEM	India	PU, PEOU, system quality, connectivity, discomfort, task technology fit, and structural assurance significantly impact the usage intention, wherein perceived monetary value, absorptive capacity, and personal innovativeness be insignificant.
Thakur, R. and Srivastava, M. (2014)	TAM, UTAUT	PU, PEOU, facilitating conditions, social influence, Security risk, PrR, monetary risk	PR, AR, personal innovativeness, behavioral intention	CB-SEM	India	AR is a significant factor in the intention to use mobile payments. Further, it ascertains that personal innovativeness affects AR, whereas AR, personal innovativeness, and PR collectively affect usage intention.

4. Data analysis and results

4.1. Exploratory factor analysis

We conducted an exploratory factor analysis (EFA) using Principal Component Analysis (PCA) with varimax rotations. Herein, thirty-three items were extracted under seven factors, accounting for 58.52% of the total variance (See Appendix 1). The items were extracted to their respective factors; seven items with low loadings (<0.5) were dropped from the analysis (Nunnally, 1994). The Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was 0.9, and Barlett's Test of Sphericity ($\chi^2 = 4492.46$; df = 780; p = 0.00) validated a sufficient sample for factor analysis (Hair et al., 2010; Kaiser, 1974). Table 3 presents the factors extracted and their description. It may be noted that lifestyle, speed and convenience are normally used as a single construct in several earlier studies. However, EFA of our data provided three distinct factors and they are, hence, labelled.

4.2. Reliability and validity

Internal consistency was measured using Cronbach's alpha; the reliability of the scales ranged from 0.80 to 0.90, indicating that the

Table 2
Demographic profile of respondents.

Characteristics	Frequency	Percentage %
Gender		
Male	227	63
Female	133	37
Age (Years)		
21 to 25	162	45
26 to 30	155	43
31 to 35	25	7
36 to 40	4	1
41 to 45	14	4
Occupation		
Salaried	166	46
Business	156	43
Student	25	7
Homemaker	13	4
Monthly Household Income (In INR)		
Below 50000	302	84
Above 50000	58	16
Category Usage of m-payments		
Movie/Air/Train Tickets	124	34
Banking Services	178	50
Insurance	5	1
Money Transfer	8	2
Utilities	45	13

Table 3
Description of factors.

Factors	Description
Lifestyle Governing (LG)	The first factor essentially brings together several variables which fit their lifestyle or govern it. At the same time, the customers feel relaxed as they do not want to stand in long queues to make payments. They prefer cashless transactions and can make instant payments that fit his/her lifestyle as they like to shop in that manner.
Security and Legitimacy (SL)	The second factor includes variables that highlight the m-payments to be secure, including the exchange of data among legitimate users maintaining extreme confidentiality while having sufficient controls for both merchants and customers to ensure data accuracy, thereby preventing fraud. These variables relate to intended usage protection, accuracy, and data authorization.
Efficiency and Satisfaction (ES)	The variables dealing with: better control of finances, becoming a satisfied customer by executing the payment successfully, and enhancing customers' lifestyle are grouped in the third factor.
Risk related (RI)	The fourth factor includes variables regarding the volume of personal information required, which will put the customers' privacy at risk with a likelihood of increased losses.
Convenience (CO)	The fifth factor highlights that availing of m-payments is a good idea and beneficial in terms of time and place.
Speed of Transaction (ST)	The variables related to ease of transactions fall in the sixth factor.
Intention to use m-payment (IU)	In the case of the seventh factor, it may be observed that the customer intends to both make use of and endorse the m-payments system, given its benefits.

scales were highly reliable. The generally agreed-upon cut-off for Cronbach's alpha is 0.7 (Nunnally, 1994); however, some exploratory research may decrease it to 0.6 (Dash and Paul, 2021; Hair et al., 2010). Further, we assessed construct validity by checking the factor loadings of each item. The average variance extracted (AVE) values above 0.7 are generally considered 'very good, while 0.5 is acceptable (Dash and Paul, 2021; Hair et al., 2010). In our case, all the values of composite reliability (CR) were above 0.7, and AVE was above 0.5, confirming thereby the convergent validity of the constructs. Table 4 indicates the results for the reliability and validity of the scales used in the study.

4.3. Common Method Bias

In any empirical study, CMB is natural, especially when all the measures include the same set of participants. Opposing views are available to assess the same. Recently, many researchers suggested procedural improvements rather than statistical measures to address CMV issues (Baumgartner et al., 2021; MacKenzie and Podsakoff, 2012). However, popular and widely used studies recommended an unrelated marker variable approach (Hair et al., 2010; Malhotra et al., 2006). Procedural remedies were undertaken to minimize the CMB. In addition, Harman's single factor test resulted in seven factors (using factor analysis with zero rotation), and the largest variance of a factor was 35.34%. Since more than one factor emerged, and a single factor did not account for a larger variance, Harman's single factor was addressed. Furthermore, a marker variable had low correlations with all seven constructs (Williams et al., 2010), which affirmed that Common Method Bias (CMB) was minimal. Additionally, we addressed the issue of nonresponse bias using Armstrong and Overton's (1977) approach. Notably, there was no significant difference between early (first quartile of the sample) and late respondents (last quartile of the sample) on all seven constructs. Therefore, the late respondents were assumed to be non-respondents, and thus, nonresponse bias was addressed.

4.4. Results of user profiling

User segmentation is a critical factor determining electronic commerce's success (Bhatnagar and Ghose, 2004). With the help of segmentation, marketers can utilize resources efficiently and tailor-make products that suit the requirements of the target population effectively (Wiese and Humbani, 2020; Quinn, 2009). Hierarchical clustering was carried out using Ward's squared Euclidean distance interval measure, resulting in two clusters based on the agglomeration coefficient variation to identify potential segments. The percentage variation of the agglomeration schedule yielded a two-cluster solution (Table 5).

Further, k-means clustering was done using the average scores of the factors extracted like lifestyle, convenience, security, legitimacy, and transaction speed for segmentation. The segments obtained were appropriately named. The cluster analysis was done using hierarchical and k-means, suggesting that the two clusters differed in their attitudes towards m-payments. The average scores in k-mean clustering on all dimensions were below average for cluster 1; hence, they were named the Hesitant. Cluster 2 comprised respondents whose average scores on all dimensions were higher; therefore, they were named the Confident.

Table 4
Test of reliability and validity.

	Cronbach's Alpha	CR (composite reliability)	AVE (average variance extracted)	MSV (maximum shared variance)	CO	LG	SL	ES	ST	RI	IU
CO	0.872	0.874	0.583	0.413	**0.763**						
LG	0.877	0.947	0.579	0.350	0.045[#2]	**0.761**					
SL	0.789	0.892	0.580	0.307	0.552	0.087	**0.762**				
ES	0.806	0.865	0.511	0.350	0.123	0.592	0.136	**0.715**			
ST	0.719	0.789	0.512	0.195	0.032	0.382	0.156	0.442	**0.714**		
RI	0.761	0.834	0.501	0.258	0.508	0.046	0.122	0.324	0.109	**0.703**	
IU	0.723	0.874	0.505	0.370	0.608	0.262	0.306	0.098	0.214	0.293	**0.705**

Table 5
Cluster analysis.

Constructs	Cluster 1 (n = 153; 42.50%)		Cluster 2 (n = 207; 57.50%)	
	Average	Standard deviation	Average	Standard deviation
CO	4.05	0.69	4.71	0.42
IU	3.79	0.80	4.66	0.45
ST	3.23	0.96	2.77	1.14
ES	3.14	0.69	4.10	0.55
RI	3.44	0.72	3.23	0.87
LG	3.71	0.63	4.48	0.41
SL	2.96	0.55	4.02	0.52

Based on the results obtained, the authors named the clusters. Since the user profiling was based on Roger's diffusion of innovation model (1995; 1962) and the user's attitude towards m-payment usage, the naming of clusters was chosen as "The Hesitant" and "The Confident". The hesitants were the consumers who were late majority or laggards with respect to choosing the m-payment service. The confidents carried positive attitude towards m-payment service usage.

4.4.1. Cluster 1: the hesitant (42.50%)

These are skeptical and cautious individuals who take longer to get convinced about new technology. They prefer to remain neutral about most things than take sides and do not share their feelings readily. They are generally dubious about new technology's performance/reliability/security. They dislike companies tracking their activities and offering targeted ads. They cannot be easily convinced by sales promotion methods like promises of cash-back, discounts, and rebates. They fiercely protect their personal information and do not share it unless mandated. They find traditional means of doing things easy and believe in tried and tested methods rather than any novelty. As far as m-payments are concerned, they are unsure about the actual functional benefits of using them. Additionally, they are highly distrustful of the current security provisions and thereby find m-payments risky.

4.4.2. Cluster 2: the confident (57.50%)

These are confident individuals who get logically convinced about different functional benefits. They have faith in the reliability of new technology. They can easily get swayed by more discounts, rebates, and cash-back promises. They are not overly worried about keeping their personal information private. They are confident in their ability to learn and adapt to new technology. As far as m-payments are concerned, they are convinced about its functional benefits in terms of convenience and ease. They have faith in the current security provisions in the m-payment apps and, thereby, do not consider it to be risky.

4.4.3. Discriminant analysis

We further applied discriminant analysis to test the suitability of the clusters obtained and determine if the classification was appropriate. Table 6 shows that the groups displayed statistically significant different means since the p-values are less than 5% significant (0.05). Therefore, Wilk's statistic was deployed to obtain the discriminant function maximizing the quotient between the variation explained by the difference between group means and variation within these groups. Further, the security variable has the lowest Wilk's statistic; thus, it provides the most significant difference between cluster means. The statistical significance of Wilk's lambda shows that the function is significant for discriminating between the clusters (Mitchell, 1994).

Looking at Table 6, column F, it was observed that null hypotheses of equal means were rejected, and alternate hypotheses were accepted since the groups displayed different means. Notably, all the variables were found to be significant in differentiating between the groups. Wilk's lambda test statistic helped determine the discriminant function that maximizes the variance between the groups based on the difference between group means and variation between the groups. It is a test of the badness of fit. The obtained results suggest that 'security' had the most significant difference between the cluster means since it produced the lowest score for lambda.

The discriminant analysis helped identify one discriminant function (Table 7). The differences between the clusters were analyzed based on the standardized canonical discriminant function coefficients (β_1 to β_7 values for seven factors).

The discriminant function equation obtained from Table 8 is:

$$D = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7$$

$$D = \beta_0 + 0.247{*}X_1 - 0.007{*}X_2 - 0.185{*}X_3 + 0.410{*}X_4 - 0.087{*}X_5 + 0.119{*}X_6 + 0.668{*}X_7$$

Where X_1 is "convenience," X_2 is "intention to use," X_3 is "speed of transaction," X_4 is "efficiency and satisfaction," X_5 is "risk related," X_6 is "lifestyle governing," and X_7 is "security and legitimacy," and β_0 is the constant.

4.4.4. Differences between the confidents and the hesitants

The discriminant analysis was used to assess whether there were any differences between the confidents (adopters, n = 207) and hesitants (non-adopters, n = 153) of mobile payments. The discriminant function equation analyzes the individual contribution of factors that act as discriminators between the two clusters. The confidents and hesitants differ significantly on their perception of security while using m-payment methods. Satisfaction with m-payment apps and convenience also contributed mainly to distinguishing the clusters from each other. Lifestyle factors (see Appendix 1) also acted as a significant discriminator between the two clusters.

5. Discussion and implications

Mobile payment systems offer users the ease of making payments with savings of time and avoid the risk of carrying physical money. Nonetheless, the benefits will accrue only if the end users adopt mobile payments. With growing mobile device usage and internet data subscribers, the scope for adopting mobile payments in India is immense. This research adopted the theoretical lens of the TAM and UTAUT and examined the factors that induce the adoption of mobile payments in an emerging economy like India.

We collected cross-sectional data from 360 users and classified the data into two sub-groups – the hesitants and the confidents. The analysis provided evidence of significant differences among the clusters concerning convenience, intention, speed, satisfaction, risk, lifestyle, and security related to newer digitization initiatives, such as m-payments. While all the variables are significant in the current study, as corroborated in earlier works (Humbani and Wiese, 2019; Park et al., 2019a,b; Thakur and Srivastava, 2014), the risk is insignificant (Refer to Table 6). It is supported by the earlier findings (Talwar et al., 2020), suggesting

Table 6
Test of equality of means.

Constructs	Wilks' Lambda	F	df1	df2	Sig.
CO	.740	125.758	1	358	.000
IU	.674	172.964	1	358	.000
ST	.956	16.497	1	358	.000
ES	.624	215.688	1	358	.000
RI	.984	5.902	1	358	.016
LG	.644	198.331	1	358	.000
SL	.505	351.193	1	358	.000

Table 7
Canonical discriminant function.

Function	Eigenvalue	Wilks' Lambda	Chi-square	Df	Sig.
1	1.588	.386	337.090	7	.000

Table 8
Standardized canonical discriminant function coefficients.

Factors	Function
CO	.247
IU	-.007
ST	-.185
ES	.410
RI	-.087
LG	.119
SL	.668

that such a scenario could arise with time and multiple m-payment options, as technology has improved considerably. Furthermore, it has made m-payment users assured that service providers have ensured adequate privacy protection and encryption measures. However, service providers must be careful in preserving privacy and security because any compromise might permanently impair their reputation.

Furthermore, the current study brings valuable insights into the penetration of digitization initiatives and provides an accurate and deep understanding of Rogers' DOI in the context of m-payments. The article has attempted to describe the m-payments phenomenon primarily by analyzing the attributes and preferences of the adopter categories of innovators, early adopters, late majority, and laggards.

5.1. Implications for theory

Theoretically, this study augments the m-payment users' behavior intention literature by examining the predictors of using m-payment technology. Market segmentation is likely to engender significant results for marketers of Mobile Payments solutions, as reported by Thakur et al. (2014). Past studies in this area have focused on technical, technological, and user-specific characteristics and their impact on the intention to use mobile-based payment services (Upadhyay and Jahanyan, 2016). Loh et al. (2020) explored the motives behind the gradual shift in usage intentions of mobile payments from cash using the perspective of the switching intention. The current study examined the attitudes towards users' adoption in light of Roger's diffusion and adoption innovation model on mobile payments from a theoretical viewpoint. Although prior studies on m-payments have received substantial attention in the electronic commerce context, the segmentation of users/non-users regarding adopting m-payments has scarcely been examined.

This study further adds to the knowledge about technology adoption by making significant and different contributions to theory. The study indicates how DOI theory can be a useful theoretical lens for investigating issues influencing technology adoption in m-payments. While several studies have looked into challenges with m-payment service uptake, few have used DOI as the foundation for their research. When individuals consider shifting from a well-entrenched legacy system to an emerging technology like m-payment services, this study supports the importance of DOI and other factors deliberated in the study.

Additionally, existing research has essentially used IT adoption theories, such as TAM (Davis, 1989), Innovation Diffusion Theory (IDT) (Rogers, 2003), and UTAUT (Venkatesh et al., 2003) to explain a mobile user's behavior (Schierz et al., 2010; Mallat, 2007).

However, it has rarely classified the users by segmenting them and identifying the traits for user adoption of m-payments. The study highlights factors that govern m-payment usage among millennials and examines the clusters based on these factors. Furthermore, the study validated that users may be classified as the Hesitants and the Confidents, which would help marketers develop strategies to retain existing users and attract new users to their platforms. We aimed to anticipate m-payment behavior by combining mobile-based predictors into an extended TAM framework, coupled with Rogers's diffusion and adoption of innovation, motivated by a growing number of intelligent device users and mobile internet use. We believe this research adds to TAM's understanding by proposing classifying users by adopting a clustering approach based on Roger's Model and further discriminating between the two clusters. Due to the fast-paced nature of this era, the original TAM may not be sufficient to explain changing customer behavior. As a result, a combination of the two approaches has been used to predict consumers' m-payment behavior intentions.

5.2. Implications for practice

Along with the theoretical implications, the implications from this study for managers are manifold. First, it provides the various determinants of m-payment adoption that will help the decision-makers to develop their marketing strategy accordingly. Second, as the study successfully segments the customers into two segments: (confidents and hesitants), it will guide the managers to understand their profiles better. Third, our findings give an insight into the techniques that m-payment service providers would want to focus on in targeting those segments of the customers that are open and willing to adopt the services (confidents), subsequently becoming opinion leaders or influencers. It works for developing economies. Fourth, the hesitants' real concerns can be addressed better if they are known to the managers. In fact, "KYC" (know your customer) is crucial in this industry, and it should be practiced literally. Fifth, the security and legitimacy was found to be the most important discriminator. It is a well-known fact that many people are still skeptical about the security and privacy aspects of m-payment services, in this case, the hesitants. This study provides a clear insight for the managers to instill faith and belief in the customers about the same. Sixth, efficiency and satisfaction, and convenience too play a huge role in influencing the decisions of the respondents. Managers must improve the efficiency of the applications with regular upgrades and improved customer service. In any kind of service, satisfaction is crucial, and the managers must meet the expectations. M-payment service providers must ensure that feedback and corrective mechanisms must be improved to convert the hesitants into the confidents. It is generally believed that m-payment is convenient, and the managers must make it so. The providers must make it more convenient than traditional services. All glitches must be solved as soon as possible after they are identified. Finally, we can conclude that these findings could assist India's m-payment industry. Several m-payment systems have been developed since the Indian government presented its aim of encouraging and supporting digitally driven transactions. Based on our research, we hope it would help in assisting m-payment service providers in India in understanding the rationale behind m-payments acceptance vis a vis its adoption intention, which in turn seems to influence the potential customers' decisions to use m-payments.

6. Future research directions

Despite its valuable contributions, the study has its own share of limitations that provide avenues for future researchers. First, the study was limited to just one city in India, and thereby, the findings are subjected to geographical constraints. Future researchers can explore broader national and international factors to address to address generalization issues efficiently. Second, the impacts of culture and socio-economic variables can be assessed to provide a better picture. In our study, we have not addressed this issue. Third, future research could additionally test the presence of other variables that possibly influence the intention to use m-payments; for instance, consumers' experience, word of mouth, familiarity/expertise, loyalty, and user personality. Fourth, more clusters should be explored with specific characteristics that can help the service providers to target the group efficiently. Fifth, we have considered only Gen Y customers in this study and we strongly believe that there can be avenues to explore the differences between Gen Y and other generations. Finally, this study has adopted a quantitative approach. Although we had conducted a FGD during pilot study, more qualitative studies would be useful. Mixed-methods approach would

provide greater validity and acceptance of the findings.

Data availability

Data will be made available on request.

Appendix 1

Sr No	Factor Name	Major studies	Statements	Loadings
1	Lifestyle Governing (LG)	Sheth et al. (1991)	LG1: Using m-payment systems would make it easier for me to manage and make payment	0.550
		Chen and Nath (2008);	LG2: I feel relaxed while making the m-payment.	0.593
		Omigie et al. (2017);	LG3: I believe that using m-payment will improve my daily life	0.763
		Sharma et al. (2019)	LG4: I believe I will do m-payment when I must make instant payments.	0.712
			LG5: I will do m-payment when I don't want to stand in a queue to make payments.	0.781
			LG6: I will use m-payment when there is a need for cashless transactions.	0.803
			LG7: I believe that using m-payment will fit my lifestyle	0.680
			LG8: I believe that using m-payment methods is compatible with the way I like to shop	0.678
2	Security and Legitimacy (SL)	Chen and Nath (2008);	SL1: I trust m-payment systems to be secure	0.653
		Daştan and Gürler, 2016;	SL2: I believe that m-payment methods incorporate sufficient security	0.710
		Liébana-Cabanillas et al. (2018);	SL3: I believe that the data exchanged in m-payment will be restricted to legitimate users only	0.775
		Sharma et al. (2019)	SL4: I believe that m-payment methods have security controls to maintain data confidentiality	0.773
			SL5: I believe that m-payment methods have controls to prevent merchants from denying having participated in a transaction	0.693
			SL6: I believe that m-payment methods have controls to ensure the accuracy of data	0.568
			SL7: I believe that m-payment methods have security controls to prevent fraud.	0.768
			SL8: I believe that my personal information stored in the m-payment databases will be protected	0.800
			SL9: I believe that my personal information stored in the m-payment databases will be accurate	0.561
			SL10: I believe that the personal information I provide for m-payment will only be used for the purposes I authorize	0.727
3	Efficiency and Satisfaction (ES)	Chen and Nath (2008);	ES1: If I use the m-payment service regularly, I believe I would get better control of my own finances	0.688
		Sharma et al. (2019);	ES2: If I use the m-payment service regularly, I believe I would become a more satisfied customer	0.678
		Dash et al. (2021)	ES3: Using m-payment systems would improve my performance in making payments	0.653
			ES4: I believe using m-payment methods will enhance the efficiency of the service provider.	0.672
			ES5: I believe that using m-payment methods will reduce human errors.	0.652
4	Risk Related (RI)	Chen and Nath (2008);	RI1: I am concerned about the amount of personal information required for making m-payments	0.546
		Johnson et al. (2018);	RI2: I believe that using m-payment will put my privacy at risk	0.704
		Thakur and Srivastava (2014)	RI3: I believe that the overall riskiness of mobile payment systems is high	0.768
			RI4: I believe there is a high probability of losing more in using m-payment systems	0.767
5	Convenience (CO)	Kim et al. (2010);	CO1: I can use m-payment anytime and anywhere	0.710
		Daştan and Gürler, 2016;	CO2: I believe m-payment services are beneficial in terms of time and place independence	0.625
		Chakraborty et al. (2022)		
6	Speed of Transaction (ST)	Chen and Nath (2008);	ST1: I believe that using m-payment will improve the speed of transactions.	0.708
		Chakraborty et al. (2022)	ST2: I believe that using m-payment will save me time.	0.738
7	Intention to use m-payment (IU)	Venkatesh and Davis (2000); Daştan and Gürler, 2016;	IU1: Given the opportunity, I will use m-payment services.	0.550
		Liébana-Cabanillas et al. (2018);	IU2: I will strongly recommend others to use m-payment systems.	0.557
		Chakraborty et al. (2022)		

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